

Dhana Bahadur Muktan

Lecturer, Machine Learning & Data Science Researcher

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PROFILE

Early-career academic and IT lecturer with a strong background in Machine Learning, Data Science, and Computer Science education. Completed Master's degree in Information Technology (MTech IT) with research focused on crop yield prediction using advanced machine learning techniques in the context of Nepal.

Passionate about teaching, applied research, and student mentorship, with experience in delivering undergraduate IT courses, conducting lab sessions, and supervising student projects. Focused on making technical concepts practical and industry-relevant to enhance student learning and problem-solving skills.

Interested in contributing to data-driven research, artificial intelligence, machine learning, and interdisciplinary academic innovation, with a strong focus on developing real-world solutions in sustainable and intelligent systems.

EXPERIENCE

12/2024 – Present
Banepa, Kavre

BITM Incharge, *Nist College Banepa*

- Managed and coordinated the BITM program, ensuring effective academic operations, curriculum implementation, and continuous program improvement.
- Taught undergraduate courses in BITM, BCA, and BSc CSIT, including OOP with Java, Information Security, System Analysis and Design, DBMS, C Programming, and Python.
- Mentored students in academics, projects, internships, and career development while fostering an engaging and supportive learning environment.
- Collaborated with faculty, organized workshops and industry interactions, and supported quality assurance initiatives to enhance student learning and employability.

05/2023 – Present
Banepa, Kavre

Computer Teacher, *Banepa Nist School*

- Provided part-time instruction in Computer Science to +2 level students, focusing on programming fundamentals, core computer concepts, and hands-on lab activities.
- Developed and delivered well-structured lessons in line with the curriculum to support clear understanding of both theoretical knowledge and practical applications.
- Assisted students with assignments, internal evaluations, and exam preparation to enhance their academic performance and confidence.
- Fostered an interactive and supportive classroom environment that encouraged logical reasoning and strengthened problem-solving abilities.

10/2020 – 12/2024
Banepa, Kavre

Computer Teacher, *Baylor International Academy*

- Delivered part-time Computer Science instruction for +2 level students, covering programming basics, computer fundamentals, and practical lab sessions.
- Prepared and delivered structured lessons aligned with the curriculum to ensure effective understanding of theoretical and practical concepts.
- Guided students through assignments, internal assessments, and exam preparation to improve academic performance.

- Maintained an engaging and supportive learning environment, promoting logical thinking and problem-solving skills.

09/2021 – 10/2022
Banepa, Kavre

Web Developer, Emerge Infosys Pvt. Ltd.

- Developed and maintained diverse web development projects, ensuring high performance and responsiveness to client needs.
- Collaborated closely with clients to deliver customized technical solutions, earning formal recognition for exemplary work and professional delivery.
- Took ownership of existing codebases, performing regular updates and optimizations to ensure system stability and security.
- Maintained a professional and courteous attitude within cross-functional teams, consistently meeting project milestones and quality standards.

EDUCATION

12/2022 – 11/2025
Dhulikhel, Kavre

Masters of Technology in Information Technology, Kathmandu University

- CGPA: 3.65
- Thesis Title: "A Comparative Study of Machine Learning Models for Cereal Crop Yield Prediction in Nepal"

09/2014 – 12/2020
Banepa, Kavre

Bachelors of Science in Computer Science and Information Technology (BSc. CSIT), NIST College Banepa (Affiliated to TU)

- Percentage: 77.79 %
- Division: First

06/2014 – 07/2016
Banepa, Kavre

Higher Secondary Board Examination (+2 Science), Banepa Nist Higher Secondary School

- Percentage: 74.50 %
- Division: First

04/2004 – 03/2014
Dapcha, Kavre

School Leaving Certificate Examination (SLC), Shree Krishna Higher Secondary School

- Percentage: 82.25 %
- Division: Distinction

SKILLS

Data Science & Analytics

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Data Preprocessing & Cleaning (Pandas, NumPy), Exploratory Data Analysis (EDA), Data Visualization (Matplotlib, Seaborn), Statistical Analysis, Dataset Handling

Programming & Software Development

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Python, SQL, Object-Oriented Programming (Java/Python), Streamlit, FastAPI

Tools & Technologies

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Git & GitHub, Google Colab / Jupyter Notebook, VS Code, MS Office, Latex, Kaggle Platform

Machine Learning & Artificial Intelligence

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Scikit-learn, Regression & Classification Models, Model Evaluation, Feature Engineering, Deep Learning, Predictive Modeling

Research & Academic Skills

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Academic Writing (Thesis & Reports), Research Design & Methodology Literature Review & Paper Structuring, Comparative Model Analysis, Problem Formulation in Applied AI

Teaching & Academic Skills

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Undergraduate Lecture Delivery, Curriculum Delivery & Lab Instruction, Student Mentoring & Supervision, Assessment Design (Assignments, Exams, Labs), Workshop Facilitation

LANGUAGE

- English
- Nepali
- Hindi
- Tamang

TRAINING/CERTIFICATIONS

12/2025 – Present **Data Science and AI Engineering** [↗](#), *Vrit Jobs*
Kathmandu, Nepal

AWARDS

25/08/2018 **The Bronze Standard**, *The Duke of Edinburgh's International Award*

- Completed 6 months of disciplined activity across service, physical, and skill-based goals.
- Led a 2-day team journey, mastering navigation and collaborative problem-solving.
- Achieved measurable technical growth while delivering consistent community service.

PROJECTS

Nepali Language Corpus Collection and Processing for Information Retrieval and NLP Applications

- Collected and curated diverse Nepali language corpora from multiple sources for NLP research, information retrieval, and language processing tasks.
- Processing and organizing datasets for tasks such as text cleaning, normalization, tokenization, and preparation for downstream machine learning applications.
- Ongoing work focused on improving dataset quality, expanding corpus coverage, and building a structured resource for Nepali language-based information retrieval systems.

A Comparative Study of Machine Learning Models for Cereal Crop Yield Prediction in Nepal [↗](#)

- Built an end-to-end ML system for cereal crop yield prediction in Nepal using agricultural, climate, soil, and fertilizer data across multiple cereal crops.
- Performed data preprocessing, feature engineering, and EDA to clean data and extract key yield-driving factors.
- Implemented and compared multiple models (Linear Regression, Decision Tree, Random Forest, Gradient Boosting, XGBoost).
- Evaluated and optimized models using MAE, RMSE, R^2 score, and cross-validation for better accuracy and generalization.
- Developed a modular ML pipeline using Python (Scikit-learn, Pandas, NumPy) for training and prediction workflow.

PUBLICATIONS

A Comparative Study of Machine Learning Models for Cereal Crop Yield Prediction of Nepal

- Research paper based on the thesis work on above topic is currently in progress.

RESEARCH INTERESTS

- Machine Learning & Predictive Modeling
- Applied Artificial Intelligence for Real-World Problems in Developing Countries
- Natural Language Processing for Low-Resource Languages (e.g., Nepali)
- Agentic AI Systems and Autonomous AI Agents
- Generative AI (LLMs, Diffusion Models, Prompt Engineering)

References

Er. Santosh Khanal, *Assistant Professor, Acting Head*, Kathmandu University
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Mr. Samish Shrestha, *Undergraduate Program Coordinator*, NIST College
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Er. Rupak Gyawali, *AI Data Specialist*, RWS Group
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